

[Aim: 100/100 in Maths]

आरंभ

① ~~BPT~~ ✓
② Similarity

TRIANGLES

Lecture #3

9th

Congruency (Photocopy)

Same shape
Same size } Congruent

10th

Similarity

Same shape
size may differ



circle



circle

→ Congruent? $\times$ shape $\checkmark$ size $\times$
→ Similar? $\checkmark\checkmark$



square



square

→ Congruent? $\checkmark\checkmark$
→ Similar? $\checkmark$

shape $\checkmark$
size $\checkmark$

★ ★ ★ ★ ★ All congruent figures are similar.
Ⓟ BUT all similar figures may or may not be congruent.

#Similarity:

- ↳ Figures are said to be similar when
 - ① They both have same shape
 - ② Size may differ.

#Criteria for Similarity of two triangles:

If $\triangle ABC \sim \triangle PRQ$

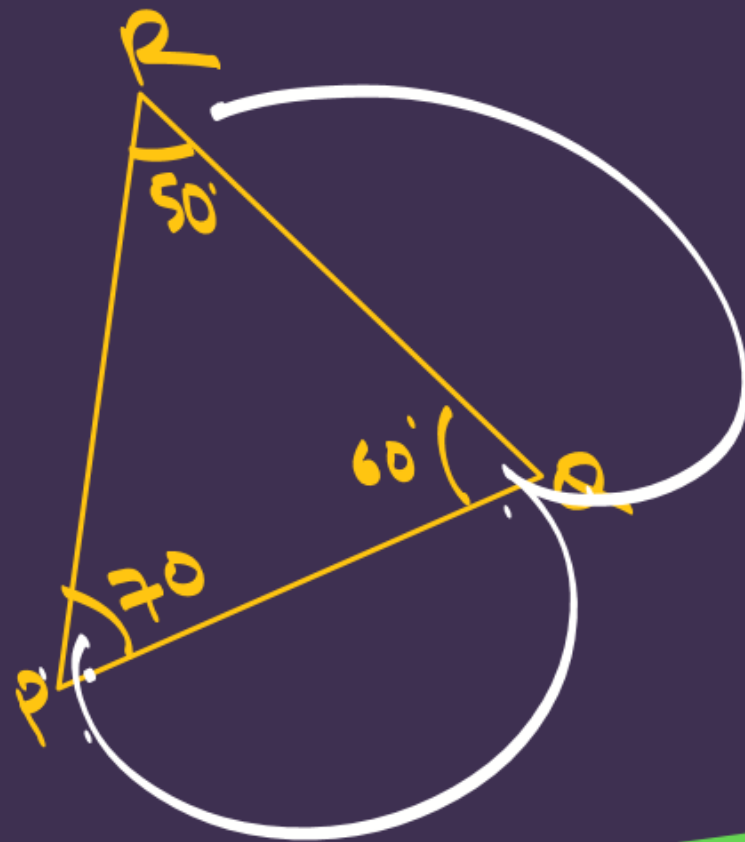
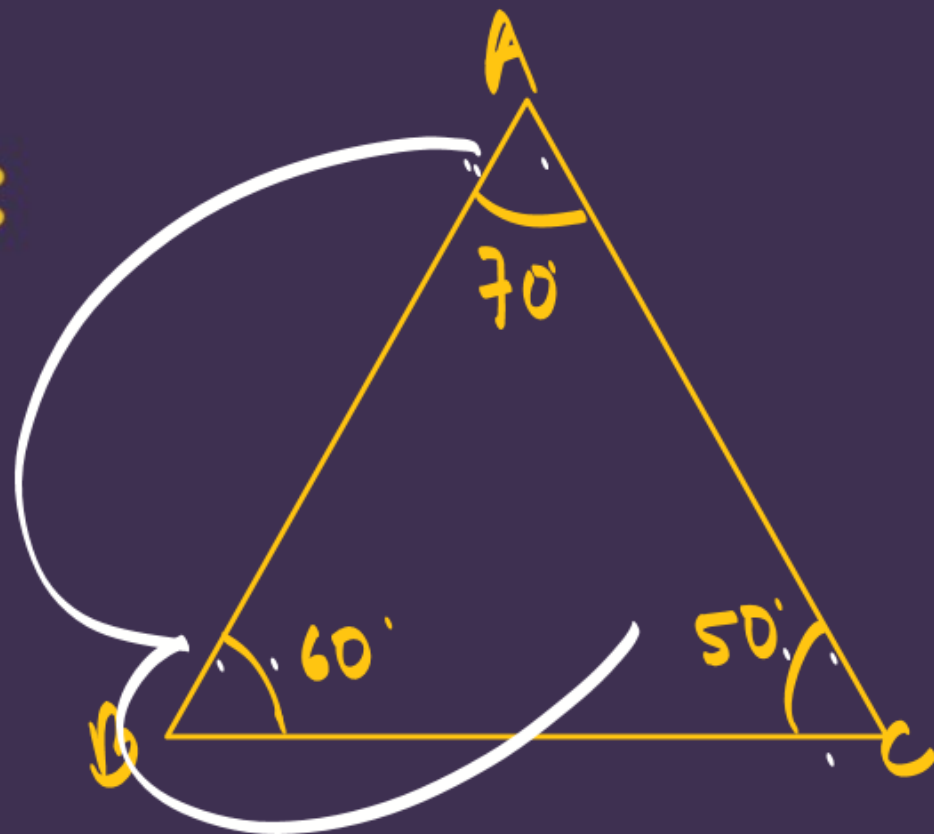
CPST { $\begin{cases} \angle A = \angle P \\ \angle B = \angle R \\ \angle C = \angle Q \end{cases}$ $\frac{AB}{PR} = \frac{BC}{RQ} = \frac{AC}{PQ}$

to prove two Δ s are similar:-

Criteria

- ~~① AAA~~
- ~~② AA~~
- ~~③ SAS~~
- ~~④ SSS~~

#1. AAA:



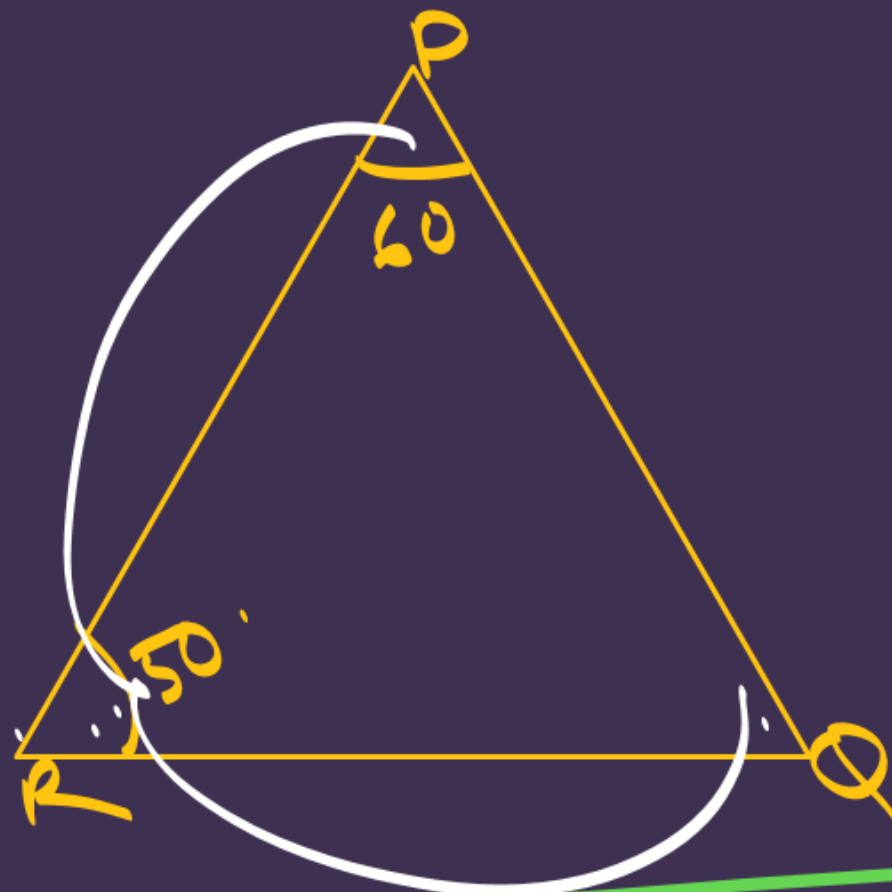
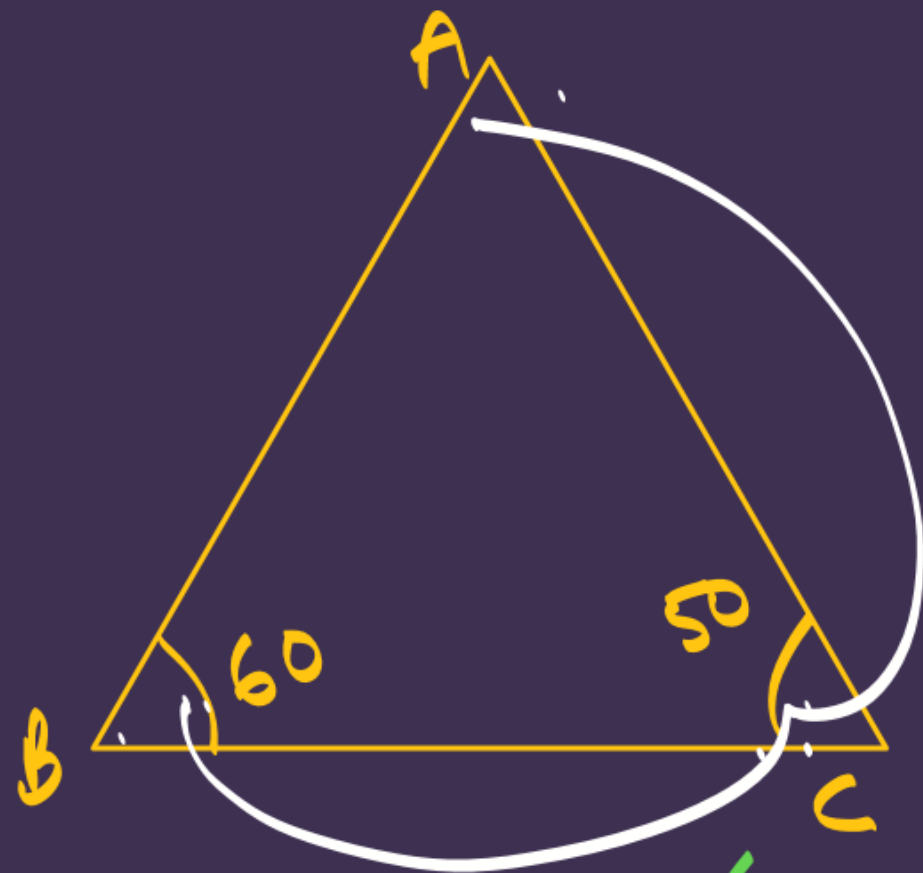
$$\begin{aligned} \angle A &= \angle P \checkmark \\ \angle B &= \angle Q \checkmark \\ \angle C &= \angle R \checkmark \end{aligned} \rightarrow$$

✓
AAA

$$\Delta \overset{\cdot}{\overset{\times}{A}}\overset{\cdot}{\overset{\times}{B}}\overset{\checkmark}{\overset{\times}{C}} \sim \Delta \overset{\cdot}{\overset{\times}{P}}\overset{\cdot}{\overset{\times}{Q}}\overset{\checkmark}{\overset{\times}{R}}$$

→

#2. AA:



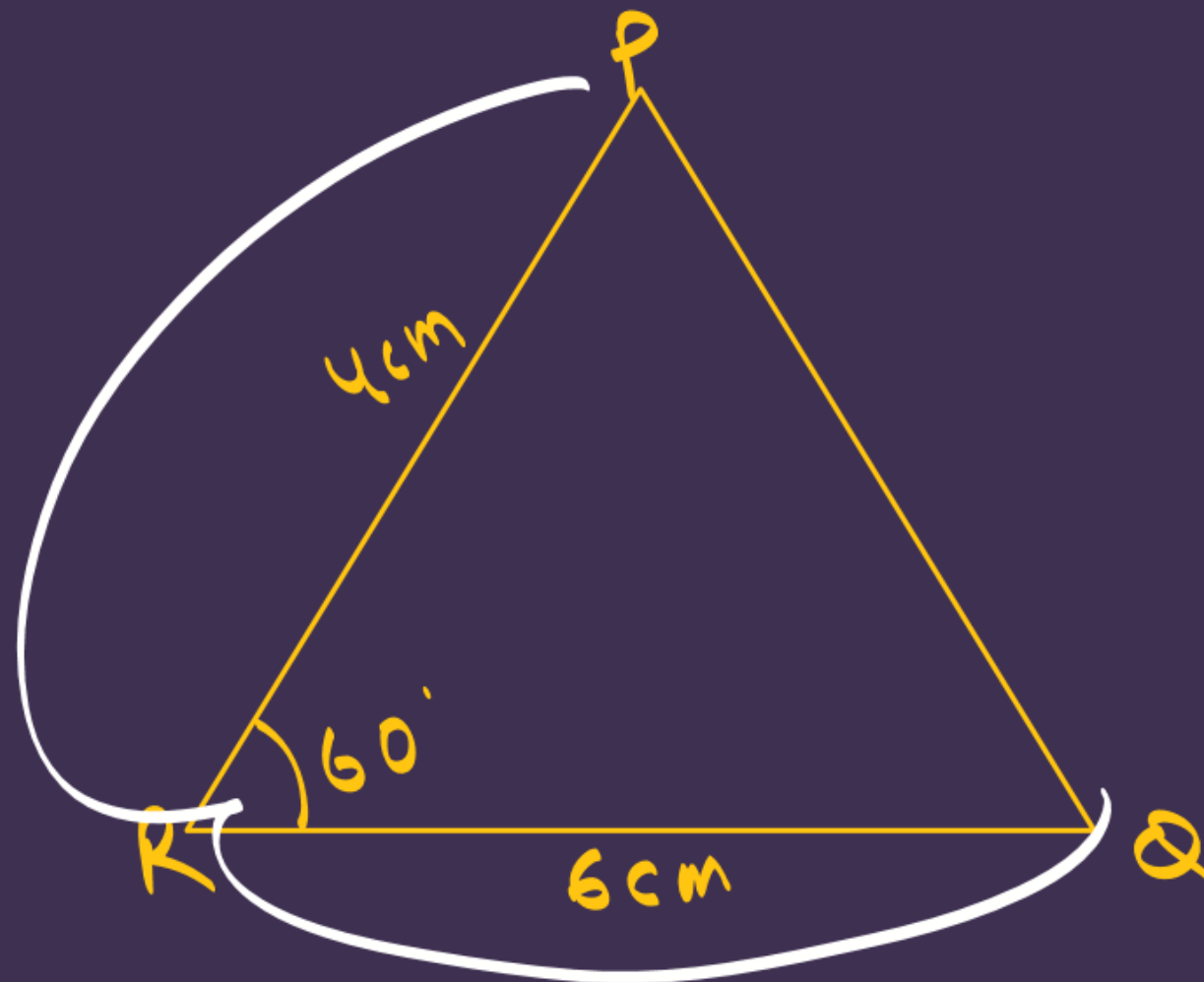
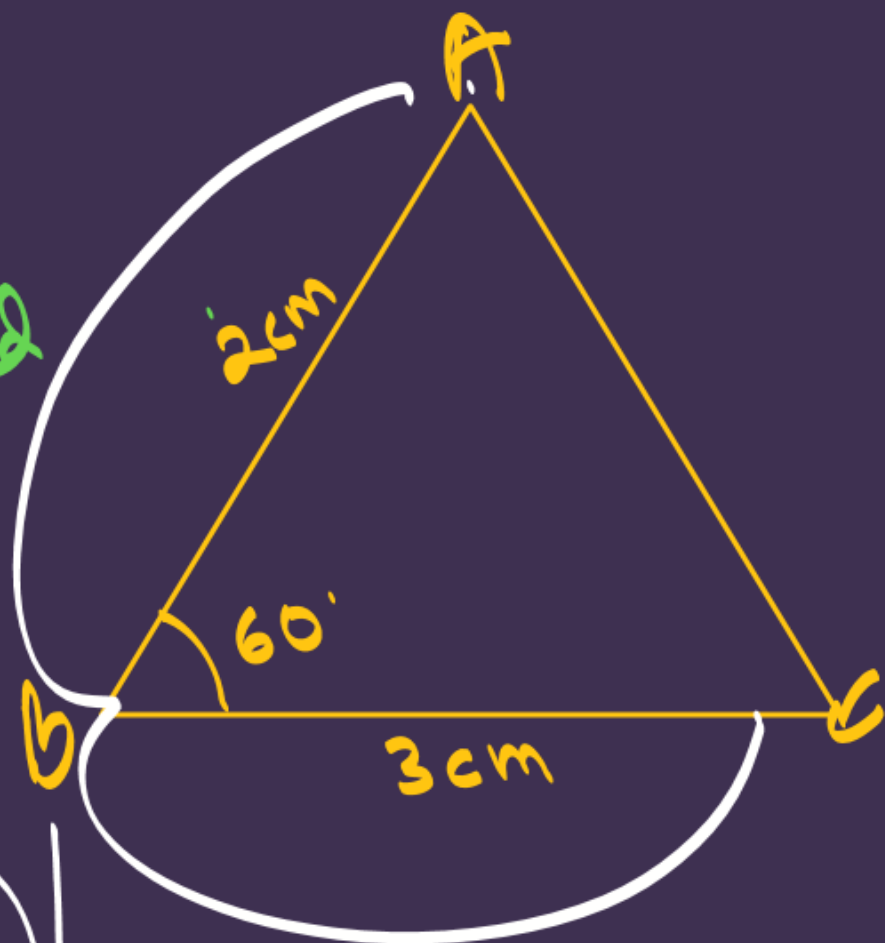
$$\begin{array}{l} \angle B = \angle P \\ \angle C = \angle R \end{array} \rightarrow$$

AA

$$\Delta BCA \sim \Delta PRQ$$

#3. SAS:

④ included angle



$$\frac{AB}{PR} = \frac{2}{4} = \frac{1}{2}$$

$$\frac{BC}{RQ} = \frac{3}{6} = \frac{1}{2}$$

$$\angle B = \angle R$$

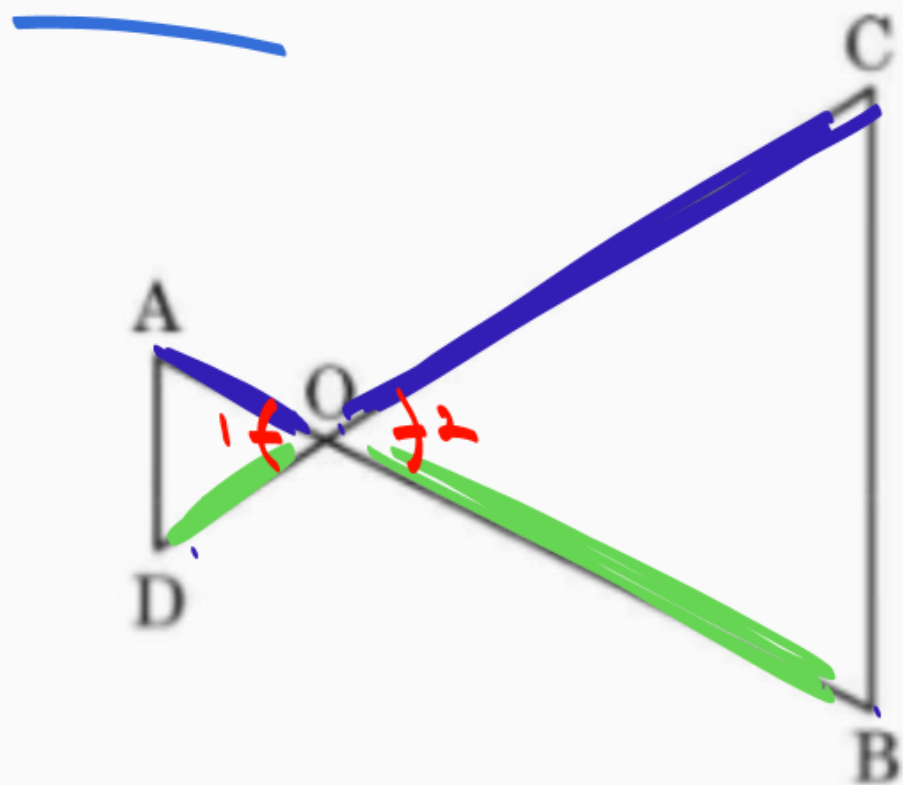
$$\frac{AB}{PR} = \frac{BC}{RQ}$$

SAS

$$\Delta ABC \sim \Delta PRQ$$

#LP:

In the given figure, $OA \cdot OB = OC \cdot OD$. Show that $\angle A = \angle C$ and $\angle B = \angle D$.



$$OA \cdot OB = OC \cdot OD$$

$$\frac{OA}{OC} = \frac{OD}{OB}$$

$$\angle 1 = \angle 2 \text{ (V.O.A.)}$$

$$\text{SAS} \Rightarrow \triangle AOD \sim \triangle COB$$

[CBSE 2025]

CPST

→ corresponding part of
Similar triangles

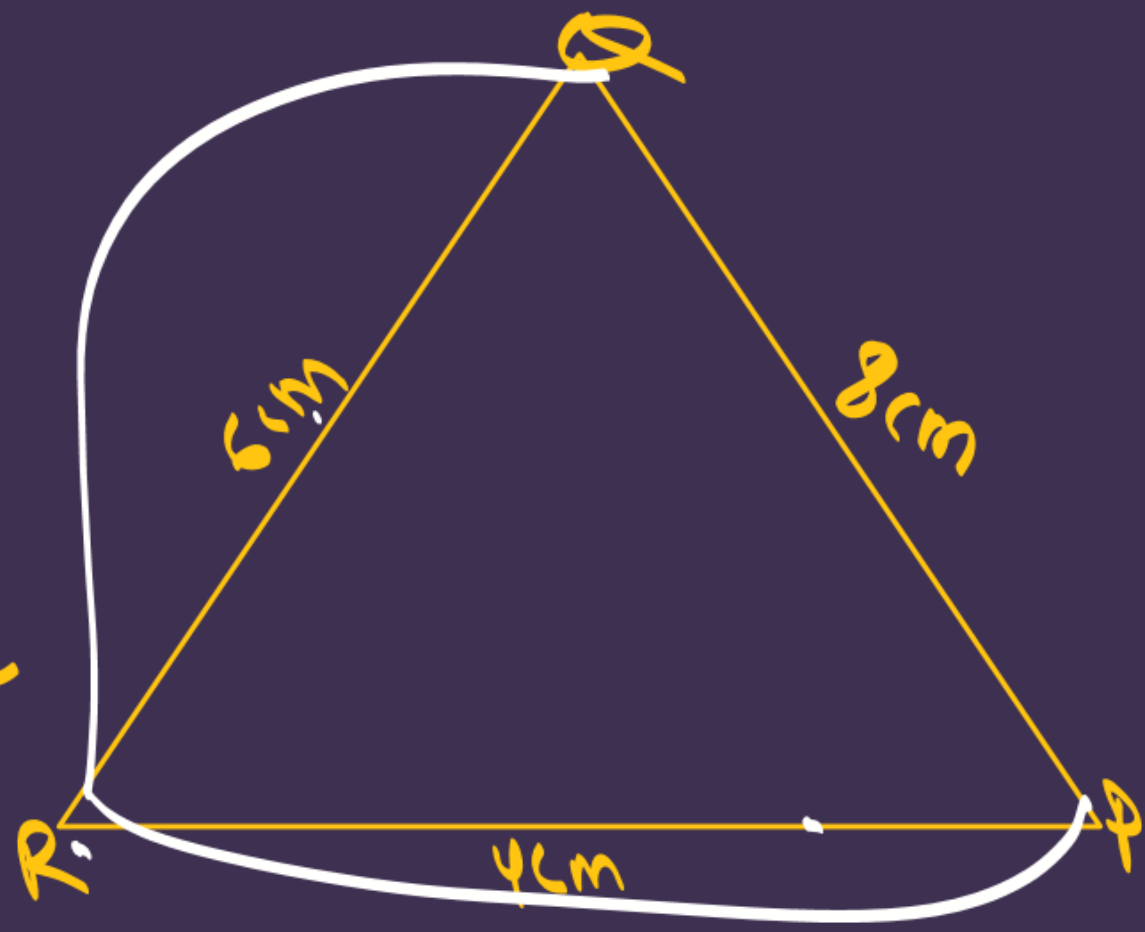
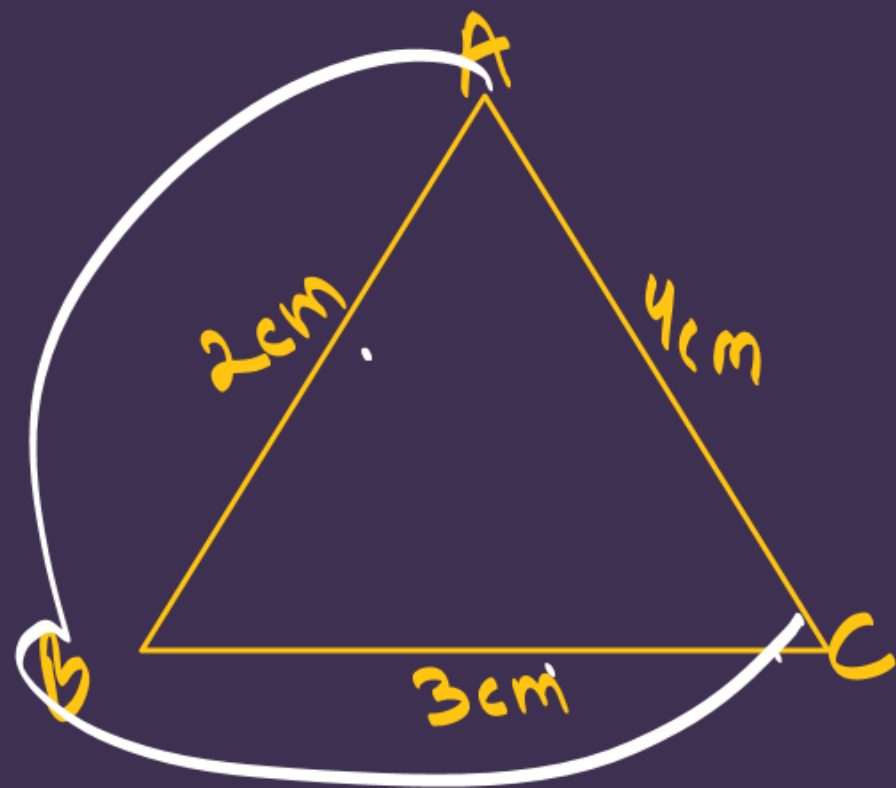
CPST ⇒

$$\angle A = \angle C$$

$$\angle D = \angle B$$

HP

#4. **SSS**:



$$\frac{AB}{RP} = \frac{2}{4} \Rightarrow \textcircled{\frac{1}{2}}$$

$$\frac{BC}{QR} = \frac{3}{6} \Rightarrow \textcircled{\frac{1}{2}}$$

$$\frac{AC}{QP} = \frac{4}{8} \Rightarrow \textcircled{\frac{1}{2}}$$

$$\left. \frac{AB}{RP} = \frac{BC}{QR} = \frac{AC}{QP} \right\} \rightarrow \boxed{SSS}$$

$$\boxed{\triangle ABC \sim \triangle PQR}$$

#LP: In two triangles ABC and PQR, $AB/QR = BC/PR = CA/PQ$, then

A. $\triangle PQR \sim \triangle CAB$

B. $\triangle PQR \sim \triangle ABC$

C. $\triangle CBA \sim \triangle PQR$

D. $\triangle BCA \sim \triangle PQR$

$$\frac{AB}{QR} = \frac{BC}{PR} = \frac{CA}{PQ}$$

#LP: If $\triangle ABC \sim \triangle PQR$, $A=30$ and $Q=90$. The value of $(R+B)$ is

A. 90

☒ B. 150

C. 20

D. 160

→ balle balle !!

cbst

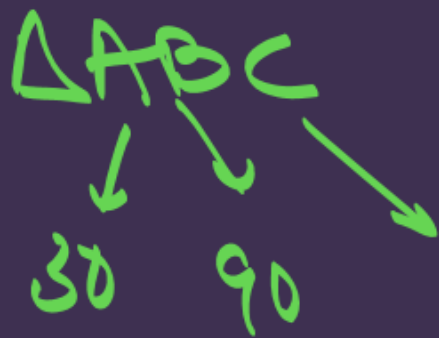
$$\angle A = \angle P = 30^\circ$$

$$\angle B = \angle Q = 90^\circ$$

$$\angle C = \angle R = 60^\circ$$

$$60^\circ + 90^\circ$$

$$\Rightarrow \boxed{150^\circ}$$



$$\text{ASP} \Rightarrow 30 + 90 + \angle C = 180$$

$$\angle C = 60^\circ$$

Thank you Goodies !!

